

Chap 1.

question 2. verify that "momentum per unit area per unit time" has the same dimensions as "force per unit area"

• momentum per unit area $(= \frac{m}{A} \cdot v)$

$$(\text{mass flux}) \times (v)$$

$$= \rho v \cdot v = \rho v^2$$

• force per unit area $(= \frac{m}{A} \cdot a)$

$$(\text{mass flux}) \times a = \rho v \cdot a$$

[momentum per unit area per unit time]

$$= \frac{\rho v^2}{dt} = \rho v \frac{d}{dt}(v)$$

$$= \rho v a = [\text{force per unit area}]$$

$\frac{d}{dt}(v) = a$, so they have the same dimensions.

question 3. compare and contrast the molecular and convective mechanisms for momentum transport.

molecular momentum flux: $\pi = p\delta + \tau$

convective momentum flux = $\rho v v$

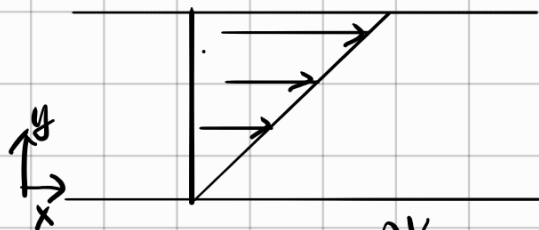
In the molecular mechanisms, the momentum flux is handling how unit area is exerted by the forces (like viscosity, or pressure) between molecules.

The force is described by dividing the unit area, and then that is the molecular momentum flux.

On the other hand, the convective momentum flux indicates another aspect of the transport by the bulk flow, and the multiplication between mass flux and velocity make the convective momentum flux.

13.1. $b > 0$

a) $v_x = by, v_y = 0, v_z = 0$



$$\nabla v = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Delta = -\mu(\nabla^2 v + (\nabla \cdot v))$$

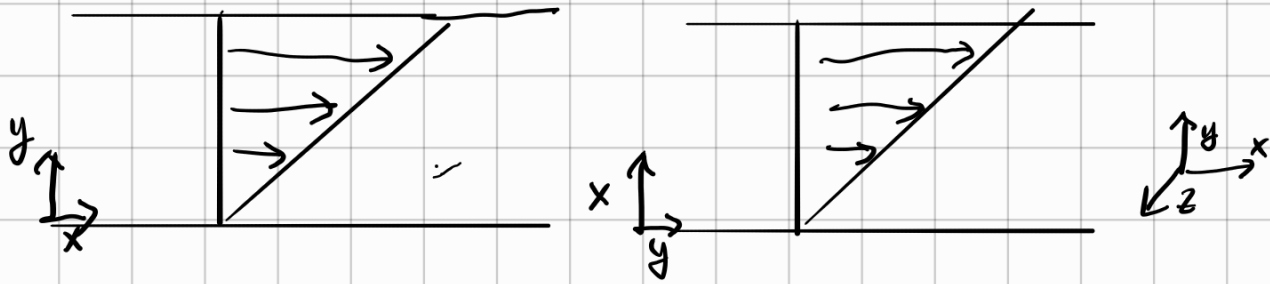
$$= -\mu \left(\begin{bmatrix} 0 & b & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right)$$

$$= \begin{bmatrix} 0 & -\mu b & 0 \\ -\mu b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho v v = \rho \begin{bmatrix} v_x & v_y & v_z \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix} = \rho \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_y v_x & v_y v_y & v_y v_z \\ v_z v_x & v_z v_y & v_z v_z \end{bmatrix}$$

$$= \rho \begin{bmatrix} (by)^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} \rho (by)^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$b) v_x = by \quad v_y = bx \quad v_z = 0$$



$$\nabla v = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

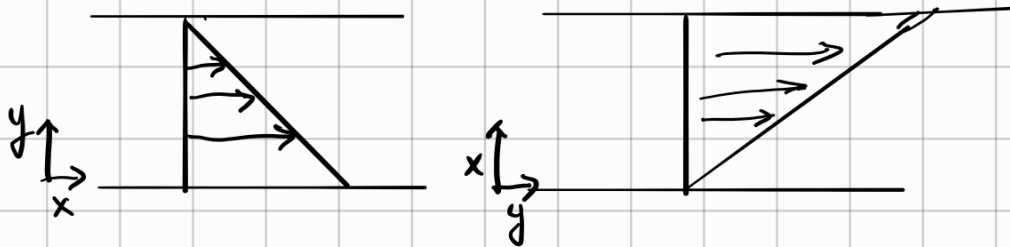
$$\tau = -\mu (\nabla v + (\nabla v)^T) = -\mu \left(\begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right)$$

$$= -\mu \begin{bmatrix} 0 & 2b & 0 \\ 2b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -2\mu b & 0 \\ -2\mu b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho v v = \rho \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_y v_x & v_y v_y & v_y v_z \\ v_z v_x & v_z v_y & v_z v_z \end{bmatrix} = \rho \begin{bmatrix} (by)^2 & (by)(bx) & 0 \\ (by)(bx) & (bx)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \rho b^2 y^2 & \rho b^2 xy & 0 \\ \rho b^2 xy & \rho b^2 x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$c) v_x = -by \quad v_y = bx \quad v_z = 0$$



$$\nabla v = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\tau = -\mu \left(\begin{bmatrix} 0 & -b & 0 \\ b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & b & 0 \\ -b & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right)$$

$$= -\mu \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\rho v v = \rho \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_y v_x & v_y v_y & v_y v_z \\ v_z v_x & v_z v_y & v_z v_z \end{bmatrix} = \rho \begin{bmatrix} (-by)^2 & (-by)(bx) & 0 \\ (bx)(-by) & (bx)^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \rho b^2 y^2 & -\rho b^2 xy & 0 \\ -\rho b^2 xy & \rho b^2 x^2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$a) v_x = -\frac{1}{2}bx \quad v_y = -\frac{1}{2}by \quad v_z = bz$$

$$\nabla v = \begin{bmatrix} \frac{\partial v_x}{\partial x} & \frac{\partial v_x}{\partial y} & \frac{\partial v_x}{\partial z} \\ \frac{\partial v_y}{\partial x} & \frac{\partial v_y}{\partial y} & \frac{\partial v_y}{\partial z} \\ \frac{\partial v_z}{\partial x} & \frac{\partial v_z}{\partial y} & \frac{\partial v_z}{\partial z} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2}b & 0 & 0 \\ 0 & -\frac{1}{2}b & 0 \\ 0 & 0 & b \end{bmatrix}$$

$$\tau = -\mu (\nabla v) + (\nabla v)^T = -\mu \left(\begin{bmatrix} -\frac{1}{2}b & 0 & 0 \\ 0 & -\frac{1}{2}b & 0 \\ 0 & 0 & b \end{bmatrix} + \begin{bmatrix} -\frac{1}{2}b & 0 & 0 \\ 0 & -\frac{1}{2}b & 0 \\ 0 & 0 & b \end{bmatrix} \right)$$

$$= -\mu \begin{bmatrix} -b & 0 & 0 \\ 0 & -b & 0 \\ 0 & 0 & 2b \end{bmatrix} = \begin{bmatrix} \mu b & 0 & 0 \\ 0 & \mu b & 0 \\ 0 & 0 & -2\mu b \end{bmatrix}$$

$$\rho v v = \rho \begin{bmatrix} v_x v_x & v_x v_y & v_x v_z \\ v_y v_x & v_y v_y & v_y v_z \\ v_z v_x & v_z v_y & v_z v_z \end{bmatrix} = \rho \begin{bmatrix} (-\frac{1}{2}bx)^2 & (-\frac{1}{2}bx)(-\frac{1}{2}by) & (-\frac{1}{2}bx)(bz) \\ (-\frac{1}{2}by)(-\frac{1}{2}bx) & (-\frac{1}{2}by)^2 & (-\frac{1}{2}by)(bz) \\ (bz)(-\frac{1}{2}bx) & (bz)(-\frac{1}{2}by) & (bz)^2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{4}\rho b^2 x^2 & \frac{1}{4}\rho b^2 xy & -\frac{1}{2}\rho b^2 xz \\ \frac{1}{4}\rho b^2 xy & \frac{1}{4}\rho b^2 y^2 & -\frac{1}{2}\rho b^2 yz \\ -\frac{1}{2}\rho b^2 xz & -\frac{1}{2}\rho b^2 yz & \rho b^2 z^2 \end{bmatrix}$$

Chap 2.

3. How are the Reynolds numbers defined for films, tubes, and spheres? What are the dimensions of Re ?

films: $Re = 4S \langle v_z \rangle \rho / \mu$

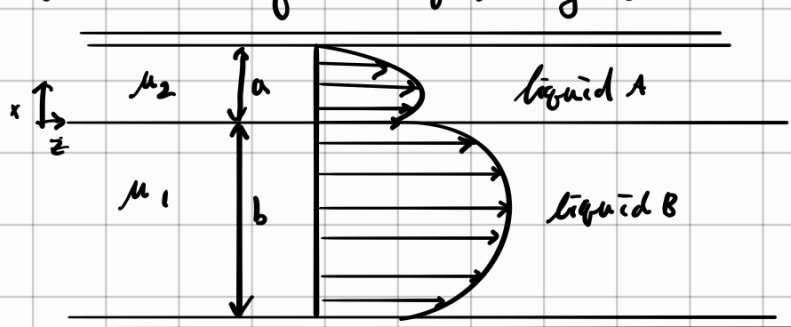
tubes: $Re = D \langle v_z \rangle \rho / \mu = 2R \langle v_z \rangle \rho / \mu$

spheres: $Re = D v_{\infty} \rho / \mu$

$$Re [=] \frac{\cancel{m} \cdot \cancel{m} / \cancel{s} \cdot \cancel{kg} / \cancel{m}^3}{\cancel{kg} / \cancel{m} \cdot \cancel{s}} = 1$$

$\therefore Re$ is a dimensionless quantity.

11. Two immiscible liquids, A and B, are flowing in laminar flow between two parallel plates. Is it possible that the velocity profiles would be of the following form? Explain



$$\begin{aligned} \mu_1 &< \mu_2 \\ a &< b \end{aligned}$$

(because density $A < B$)

$x=0, \quad v_{z,2} = v_{z,1}$

$x=-b, \quad v_{z,1} = 0$

$x=+a, \quad v_{z,2} = 0$

$$v_{z,1} = -\left(\frac{P_0 - P_L}{2\mu_1 L}\right) x^2 - \frac{C_1}{\mu_1} x + C_{2,1} \quad \frac{1}{2\mu_1} >$$

$$v_{z,2} = -\left(\frac{P_0 - P_L}{2\mu_2 L}\right) x^2 - \frac{C_1}{\mu_2} x + C_{2,2}$$

① $C_{2,1} = C_{2,2}$

② $-\left(\frac{P_0 - P_L}{2\mu_1 L}\right)(-b)^2 - \frac{C_1}{\mu_1}(-b) + C_{2,1} = 0$

$-\left(\frac{P_0 - P_L}{2\mu_2 L}\right)(a)^2 - \frac{C_1}{\mu_2}(a) + C_{2,2} = 0$

$$-\left(\frac{P_0 - P_L}{2\mu_1 L}\right) b^2 - \frac{C_1}{\mu_1}(-b) = -\left(\frac{P_0 - P_L}{2\mu_2 L}\right) a^2 - \frac{C_1}{\mu_2}(a)$$

$$\left(\frac{a}{\mu_2} + \frac{b}{\mu_1}\right) C_1 = \left(-\frac{a^2}{\mu_2} + \frac{b^2}{\mu_1}\right) \left(\frac{P_0 - P_L}{2L}\right)$$

$$C_1 = \left(\frac{-a^2\mu_1 + b^2\mu_2}{\cancel{\mu_2\mu_1} (a\mu_1 + b\mu_2)} \right) \left(\frac{P_0 - P_L}{2L} \right)$$

$$= \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) \left(\frac{P_0 - P_L}{2L} \right)$$

$$C_{2,1} = \left(\frac{P_0 - P_L}{2\mu_1 L} \right) b^2 - \frac{1}{\mu_1} \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) \left(\frac{P_0 - P_L}{2L} \right)$$

$$= \left(\frac{P_0 - P_L}{2\mu_1 L} \right) \left(b^2 - \frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right)$$

$$C_{2,2} = \left(\frac{P_0 - P_L}{2\mu_2 L} \right) a^2 + \frac{1}{\mu_2} \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) \left(\frac{P_0 - P_L}{2L} \right)$$

$$= \left(\frac{P_0 - P_L}{2\mu_2 L} \right) \left(a^2 + \frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right)$$

$$V_{z,1} = - \left(\frac{P_0 - P_L}{2\mu_1 L} \right) x^2 - \left(\frac{P_0 - P_L}{2\mu_1 L} \right) \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) x + \left(\frac{P_0 - P_L}{2\mu_1 L} \right) \left(b^2 - \frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right)$$

$$= - \left(\frac{P_0 - P_L}{2\mu_1 L} \right) \left(x^2 + \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) x + \frac{-a^2\mu_1 + b^2\mu_2 - b^2}{a\mu_1 + b\mu_2} \right)$$

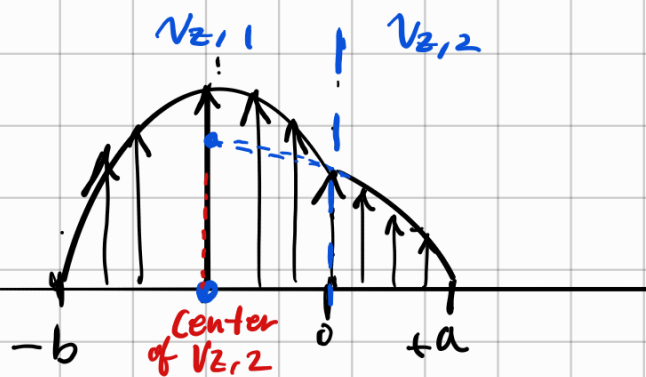
$$V_{z,2} = - \left(\frac{P_0 - P_L}{2\mu_2 L} \right) x^2 - \left(\frac{P_0 - P_L}{2\mu_2 L} \right) \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) x + \left(\frac{P_0 - P_L}{2\mu_2 L} \right) \left(a^2 + \frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right)$$

$$= - \left(\frac{P_0 - P_L}{2\mu_2 L} \right) \left(x^2 + \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) x + \frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} + a^2 \right)$$

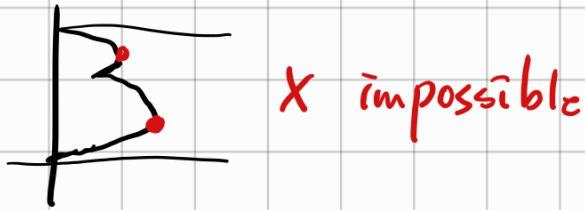
center of $V_{z,2} = -\frac{1}{2} \left(\frac{-a^2\mu_1 + b^2\mu_2}{a\mu_1 + b\mu_2} \right) < 0$

because

$$\left[\begin{array}{l} (a\mu_1 + b\mu_2) > 0 \quad (\because a, b, \mu_1, \mu_2 > 0) \\ -a^2\mu_1 + b^2\mu_2 > 0 \quad (\because a < b \text{ \& } \mu_1 \leq \mu_2) \end{array} \right]$$



∴ It is not possible because each section of A and B shows the vertex of the parabola in the following form.



2B.2.

$$i) \frac{d\tau_{xz}}{dx} = \rho g \cos \beta$$

$$\tau_{xz} = \int \rho g \cos \beta \cdot dx$$

$$= \rho g \cos \beta x$$

$$-\mu \frac{dv_z}{dx} = \rho g \cos \beta x + C_1$$

$$\frac{dv_z}{dx} = \frac{\rho g \cos \beta x + C_1}{-\mu}$$

$$v_z = \int \frac{\rho g \cos \beta}{-\mu} x + C_1 dx = \left[-\frac{\rho g \cos \beta}{2\mu} x^2 + C_1 x + C_2 \right]$$

$$ii) \frac{d\tau_{xz}}{dx} = \rho g \cos \beta \Rightarrow -\mu \frac{d^2 v_z}{(dx)^2} = \rho g \cos \beta$$

$$\frac{d^2 v_z}{(dx)^2} = -\frac{\rho g \cos \beta}{\mu}$$

$$\frac{dv_z}{dx} = \int -\frac{\rho g \cos \beta}{\mu} dx = -\frac{\rho g \cos \beta}{\mu} x + C_1$$

$$v_z = \int -\frac{\rho g \cos \beta}{\mu} x + C_1 dx = \left[-\frac{\rho g \cos \beta}{2\mu} x^2 + C_1 x + C_2 \right]$$

$$2B.3 a) \frac{\lim_{\Delta x \rightarrow 0} [W \Delta x \phi_{zz}]_0 - [W \Delta x \phi_{zz}]_L + [L W \phi_{xz}]_x - [L W \phi_{xz}]_{x+\Delta x} + W L \Delta x \rho g_z = 0}{W \Delta x L}$$

$$\lim_{\Delta x \rightarrow 0} \left(\frac{[\phi_{xz}]_{x+\Delta x} - [\phi_{xz}]_x}{\Delta x} \right) = \frac{[\phi_{zz}]_0 - [\phi_{zz}]_L}{L} + \rho g_z$$

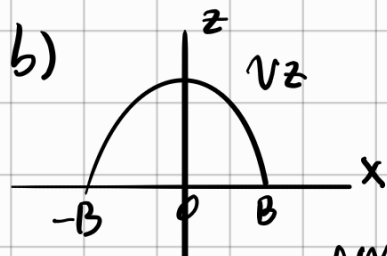
$$\frac{d\tau_{xz}}{dx} = \frac{(\rho v_x v_z + p_0 - \rho g_0) - (\rho v_x v_z + p_L - \rho g_L)}{L} = \frac{(p_0 - p_L)}{L}$$

$$\boxed{\tau_{xz} = \frac{(p_0 - p_L)}{L} x}$$

$$\frac{dv_z}{dx} = -\frac{(p_0 - p_L)}{\mu L} x, \quad v_z = -\frac{1}{2} \left(\frac{p_0 - p_L}{\mu L} \right) x^2 + C_1$$

$$x=B, v_z=0, \quad 0 = -\frac{1}{2} \left(\frac{p_0 - p_L}{\mu L} \right) B^2 + C_1, \quad C_1 = \frac{(p_0 - p_L) B^2}{2\mu L}$$

$$v_z = \frac{(p_0 - p_L) B^2}{2\mu L} \left(1 - \left(\frac{x}{B} \right)^2 \right)$$



$$v_z = \frac{(p_0 - p_L) B^2}{2\mu L} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

$$\langle v_z \rangle = \frac{\int_0^W \int_{-B}^B v_z dx dy}{\int_0^W \int_{-B}^B dx dy} = \frac{1}{2B} \cdot \frac{(p_0 - p_L) B^2}{2\mu L} \int_{-B}^B \left(1 - \left(\frac{x}{B} \right)^2 \right) dx$$

$$= \frac{1}{2B} \cdot \frac{(p_0 - p_L) B^2}{2\mu L} \left[x - \frac{1}{3B^2} x^3 \right]_{-B}^B$$

$$= \frac{1}{2B} \frac{(p_0 - p_L) B^2}{2\mu L} \left[2B - \frac{1}{3B^2} (2B^3) \right] = \frac{(p_0 - p_L) B^2}{3\mu L}$$

at $x=0$

$$v_{z,max} = \frac{(p_0 - p_L) B^2}{2\mu L}$$

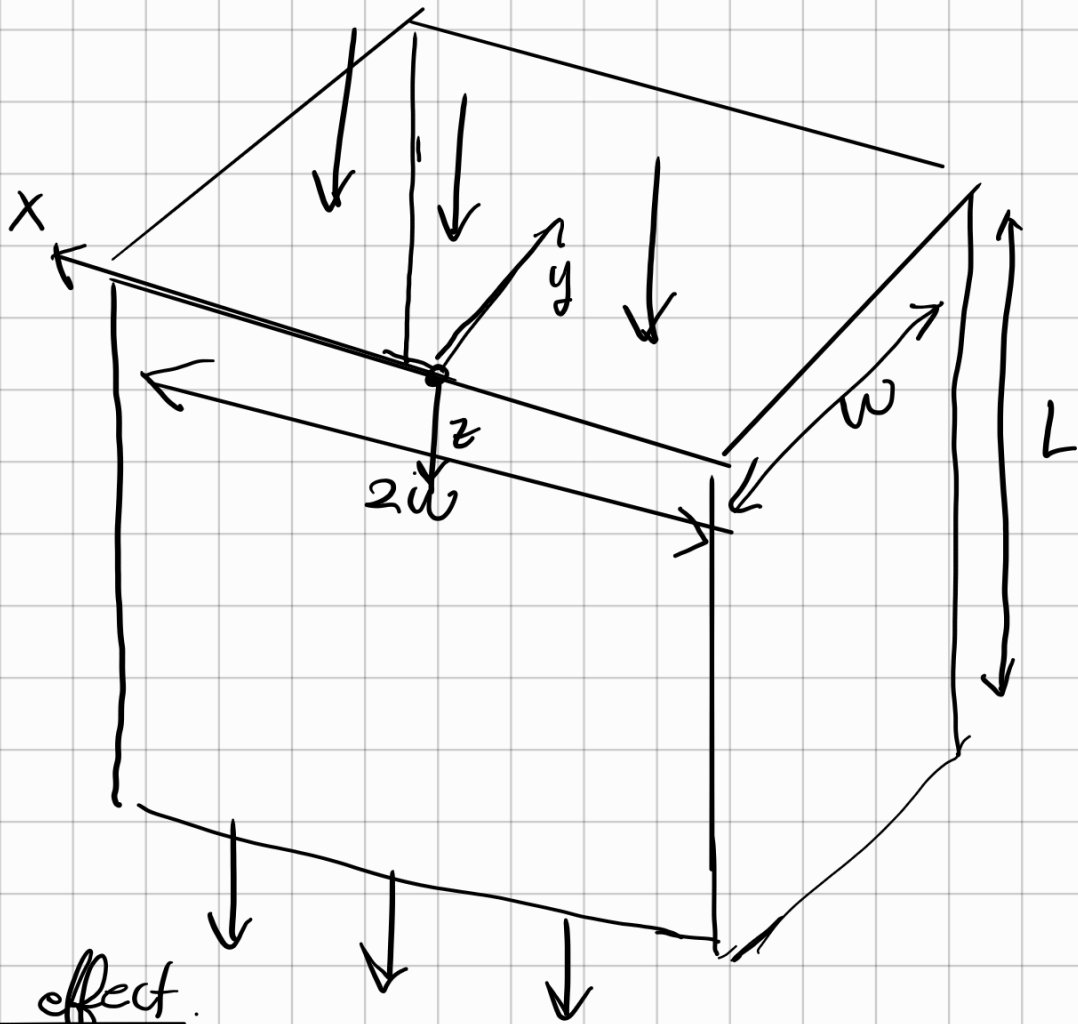
$$\frac{\langle v_z \rangle}{v_{z,max}} = \frac{\frac{(p_0 - p_L) B^2}{3\mu L}}{\frac{(p_0 - p_L) B^2}{2\mu L}} = \boxed{\frac{2}{3}}$$

$$c) W = \int_0^W \int_{-B}^B \rho v_z dx dy$$

$$= W \rho \left(\frac{(p_0 - p_L) B^2}{2\mu L} \right) \left[x - \frac{1}{3B^2} x^3 \right]_{-B}^B$$

$$= W \rho \cdot \frac{(p_0 - p_L) B^2}{2\mu L} \cdot \left(\frac{4}{3} B \right) = \boxed{\frac{2}{3} \frac{(p_0 - p_L) B^3 W \rho}{\mu L}}$$

d)



$$e) \quad \lim_{\Delta x \rightarrow 0} \frac{[w \Delta x \phi_{zz}]_0 - [w \Delta x \phi_{zz}]_L + [L w \phi_{xz}]_x - [L w \phi_{xz}]_{x+\Delta x} + w L \Delta x \rho g_z = 0}{w \Delta x L}$$

$$\lim_{\Delta x \rightarrow 0} \left(\frac{[\phi_{xz}]_{x+\Delta x} - [\phi_{xz}]_x}{\Delta x} \right) = \frac{[\phi_{zz}]_0 - [\phi_{zz}]_L}{L} + \rho g_z$$

$\phi_{xz} = -\mu \frac{\partial v_z}{\partial x} + \rho v_x v_z$
 $\phi_{zz} = p - 2\mu \frac{\partial v_z}{\partial z} + \rho v_z v_z$
 $(\because v_x = 0, \frac{\partial v_z}{\partial z} = 0)$

$$\frac{d\tau_{xz}}{dx} = \frac{(\rho v_x v_z + p_0 - \rho g_0) - (\rho v_x v_z + p_L - \rho g_L)}{L} = \frac{(p_0 - p_L)}{L}$$

$$\tau_{xz} = \frac{(p_0 - p_L)}{L} x$$

$$-\mu \frac{dv_z}{dx} = \frac{(p_0 - p_L)}{L} x \quad \frac{dv_z}{dx} = -\frac{(p_0 - p_L)}{\mu L} x$$

$$v_z = -\frac{1}{2} \frac{(p_0 - p_L)}{\mu L} x^2 + C_1$$

$$x = w, v_z = 0 \quad 0 = -\frac{1}{2} \frac{(p_0 - p_L)}{\mu L} w^2 + C_1 \quad C_1 = \frac{(p_0 - p_L) w^2}{2 \mu L}$$

$$\therefore v_z = \frac{(p_0 - p_L) w^2}{2 \mu L} \left(1 - \left(\frac{x}{w} \right)^2 \right)$$

$$2B.7 \ a) \quad r \frac{dv_z}{dr} = C_1 \quad \frac{dv_z}{dr} = \frac{C_1}{r} \quad v_z = \int \frac{C_1}{r} dr$$

$$v_z = C_1 \ln r + C_2$$

$$\left. \begin{array}{l} r = kR, \quad v_{kR} = v_0 = C_1 \ln(kR) + C_2 \\ r = R, \quad v_R = 0 = C_1 \ln(R) + C_2 \end{array} \right\} -$$

$$v_0 = C_1 (\ln(kR) - \ln(R))$$

$$C_1 = \frac{v_0}{\ln\left(\frac{kR}{R}\right)} = \frac{v_0}{\ln(k)}$$

$$C_2 = -C_1 \ln(R) = \frac{-v_0 \ln(R)}{\ln(k)}$$

$$\frac{v_z}{v_0} = \frac{\frac{v_0}{\ln(k)} \ln r - \frac{v_0 \ln(R)}{\ln(k)}}{\frac{v_0}{\ln(k)} \ln(kR) - \frac{v_0 \ln(R)}{\ln(k)}} = \frac{\cancel{\ln(k)} (\ln(r) - \ln(R))}{\cancel{\ln(k)} (\ln(kR) - \ln(R))} = \frac{\ln(r/R)}{\ln(k)}$$

$$b) \quad W = \int_{kR}^R \rho v_z 2\pi r dr$$

$$= \rho \cdot 2\pi \int_{kR}^R \left(\frac{v_0}{\ln(k)} \right) (\ln(r/R)) r dr$$

$$= \frac{\rho 2\pi v_0}{\ln(k)} \int_{kR}^R r \ln(r/R) dr \quad \left(\because \int r \ln(r/R) dr = \frac{1}{2} r^2 \ln(r/R) - \int \frac{1}{2} R^2 \cdot \frac{1}{r} dr \right)$$

$$= \frac{\rho 2\pi v_0}{\ln(k)} \left[\frac{1}{2} r^2 \ln(r/R) - \frac{1}{4} r^2 \right]_{kR}^R$$

$$= \frac{\rho 2\pi v_0}{\ln(k)} \left[\frac{1}{2} R^2 \ln(1) - \frac{1}{2} k^2 R^2 \ln(k) - \frac{1}{4} R^2 + \frac{1}{4} k^2 R^2 \right]$$

$$= \frac{\cancel{\rho} 2\pi v_0}{\cancel{\ln(k)}} \left[R^2 \cdot \frac{k^2 - 1 - 2k^2 \ln(k)}{4} \right]$$

$$= \frac{\rho \pi v_0 R^2}{2} \left[\frac{-(k^2 - 1)}{-(\ln(k))} - 2k^2 \right] = \frac{\pi R^2 \rho v_0}{2} \left[\frac{1 - k^2}{\ln(1/k)} - 2k^2 \right]$$

$$\begin{aligned}
 c) \tau_{rz} &= \mu \frac{dv_z}{dr} = \mu \frac{d}{dr} \left(\frac{v_0}{\ln(k)} \ln(r/R) \right) \\
 &= \mu \frac{d}{dr} \left[\frac{v_0}{\ln(k)} (\ln(r) - \ln(R)) \right] \\
 &= \mu \frac{v_0}{\ln(k)} \cdot \frac{1}{r} = \frac{\mu v_0}{r \ln(k)}
 \end{aligned}$$

$$r = kR,$$

$$\tau_{rz} = \frac{\mu v_0}{kR \ln(k)}$$

$$\begin{aligned}
 F_z &= \tau_{rz} (2\pi \cdot (kR) \cdot L) = \frac{\mu v_0}{kR \ln(k)} (2\pi (kR) \cdot L) \\
 &= \frac{2\pi v_0 \mu L}{\ln(k)} = - \frac{2\pi L \mu v_0}{\ln(1/k)}
 \end{aligned}$$

$$d) \varepsilon = 1 - k.$$

$$\ln(1/k) = \ln\left(\frac{1}{1-\varepsilon}\right) \approx \varepsilon + \frac{\varepsilon^2}{2} + \dots$$

$$\frac{1}{\ln(1/k)} = \frac{1}{\varepsilon + \frac{\varepsilon^2}{2} + \frac{\varepsilon^3}{3} + \dots} = \frac{1}{\varepsilon} \frac{1}{1 + \frac{\varepsilon}{2} + \frac{\varepsilon^2}{3} + \dots} = \frac{1}{\varepsilon} \cdot \frac{1}{1+x}$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \dots \quad (x = \frac{\varepsilon}{2} + \frac{\varepsilon^2}{3} + \dots)$$

$$\frac{1}{\varepsilon} \cdot (1 - x + x^2 - x^3 + \dots) = \frac{1}{\varepsilon} \left(1 - \frac{\varepsilon}{2} - \frac{\varepsilon^2}{12} + \dots \right)$$

$$F_z = - \frac{2\pi L \mu v_0}{\varepsilon} \left(1 - \frac{1}{2}\varepsilon - \frac{1}{12}\varepsilon^2 + \dots \right)$$

$$2B.8 \quad \rho = (1.594 - 0.998) \times 1000 \text{ kg/m}^3 = 596 \text{ kg/m}^3$$

$$\text{for circular tube, } w = \frac{\pi \rho R^4}{8 \mu L} \left(- \frac{dp}{dz} \right)$$

$$w = \frac{\pi (596 \text{ kg/m}^3) (1.27 \times 10^{-4} \text{ m})^4}{8 (1.0019 \times 10^3 \text{ Pa} \cdot \text{s}) (3.04 \text{ cm})} (9.81 \text{ m/s}^2) (0.0254 \text{ m}) = 4.743 \times 10^{-12} \text{ m}^3/\text{s}$$

$$(R = \frac{0.0102 \text{ m}}{2} = 1.27 \times 10^{-4} \text{ m}, L = 10 \text{ ft} = 3.048 \text{ m}, -\frac{dp}{dz} = \rho g h)$$

$$\dot{m} = \rho_w w$$

$$= (998 \text{ kg/m}^3) (4.743 \times 10^{-12} \text{ m}^3/\text{s})$$

$$= (4.734 \times 10^{-9} \text{ kg/s}) \left(\frac{2.205 \text{ lbm}}{1 \text{ kg}} \right) \left(\frac{60 \text{ s}}{1 \text{ min}} \right) \left(\frac{60 \text{ min}}{1 \text{ hr}} \right)$$

$$= 3.757 \times 10^{-5} \text{ lbm/hr}$$